

This exam has 4 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

By participating in this exam, you underwrite the following **Statement of Integrity**:

I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to make this exam completely on my own, only consult those sources that are allowed explicitly, not share my solutions with other students, and make myself available for any oral clarifications regarding this exam.

1. Equivalence relations (7+8 points)

Suppose that in the set $A := \{0, 1, 2, 3, 4, 5\} \times \{0, 1, 2, 3, 4, 5\}$ we define the following relation

$$\langle a, b \rangle R \langle c, d \rangle :\Leftrightarrow a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3},$$

where $x \equiv y \pmod{m} :\Leftrightarrow y - x$ is a multiple of m .

- a) Show that R is an equivalence relation.

Solution:

- R is reflexive: $a \equiv a \pmod{2}$ and $b \equiv b \pmod{3} \Rightarrow \langle a, b \rangle R \langle a, b \rangle$;
- R is transitive: $\langle a, b \rangle R \langle c, d \rangle$ and $\langle c, d \rangle R \langle e, f \rangle \Rightarrow a \equiv c \pmod{2}$ and $c \equiv e \pmod{2}$ and $b \equiv d \pmod{3}$ and $d \equiv f \pmod{3} \Rightarrow a \equiv e \pmod{2}$ and $b \equiv f \pmod{3} \Rightarrow \langle a, b \rangle R \langle e, f \rangle$;
- R is symmetric: $\langle a, b \rangle R \langle c, d \rangle \Rightarrow a \equiv c \pmod{2}$ and $b \equiv d \pmod{3} \Rightarrow c \equiv a \pmod{2}$ and $d \equiv b \pmod{3} \Rightarrow \langle c, d \rangle R \langle a, b \rangle$.

- b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

Solution: The different equivalence classes are

$\{\langle 0, 0 \rangle, \langle 2, 0 \rangle, \langle 4, 0 \rangle, \langle 0, 3 \rangle, \langle 2, 3 \rangle, \langle 4, 3 \rangle\}$, $\{\langle 1, 0 \rangle, \langle 3, 0 \rangle, \langle 5, 0 \rangle, \langle 1, 3 \rangle, \langle 3, 3 \rangle, \langle 5, 3 \rangle\}$,
 $\{\langle 0, 1 \rangle, \langle 2, 1 \rangle, \langle 4, 1 \rangle, \langle 0, 4 \rangle, \langle 2, 4 \rangle, \langle 4, 4 \rangle\}$, $\{\langle 1, 1 \rangle, \langle 3, 1 \rangle, \langle 5, 1 \rangle, \langle 1, 4 \rangle, \langle 3, 4 \rangle, \langle 5, 4 \rangle\}$,
 $\{\langle 0, 2 \rangle, \langle 2, 2 \rangle, \langle 4, 2 \rangle, \langle 0, 5 \rangle, \langle 2, 5 \rangle, \langle 4, 5 \rangle\}$, $\{\langle 1, 2 \rangle, \langle 3, 2 \rangle, \langle 5, 2 \rangle, \langle 1, 5 \rangle, \langle 3, 5 \rangle, \langle 5, 5 \rangle\}$;
and a complete system of representatives is given by
 $\{\langle 0, 0 \rangle, \langle 1, 0 \rangle, \langle 0, 1 \rangle, \langle 1, 1 \rangle, \langle 0, 2 \rangle, \langle 1, 2 \rangle\}$.

2. Functions and cardinality (5+5+5 points)

We are given the functions $\exp : \mathbb{R} \rightarrow \mathbb{R}$, $\text{add} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{root} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\exp(x) = 2^x, \text{add}(x) = x + 1, \text{and } \text{root}(x) = \sqrt{x}.$$

- a) For each function, determine whether or not the function is surjective.

Solution: $\exp$ is not surjective: -1 (or any $x \leq 0$) is not in the range (image);

add is surjective: for every $x \in \mathbb{R}$ we have $\text{add}(x - 1) = x$;

root is not surjective: -1 (or any $x < 0$) is not in the range (image).

- b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by $f = \exp \circ \text{root} \circ \text{add}^{-1}$.

Solution: $f(x) = (\exp \circ \text{root} \circ \text{add}^{-1})(x) = (\exp \circ \text{root})(x - 1) = \exp(\sqrt{x - 1}) = 2^{\sqrt{x - 1}}$.

- c) Express the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(x) = \sqrt{2^x} - 2$$

as a composition of $\exp$, add , root , and their inverses.

Solution: $\sqrt{2^x} - 2 = \text{add}^{-1}(\sqrt{2^x} - 1) = (\text{add}^{-1} \circ \text{add}^{-1})(\sqrt{2^x})$
 $= (\text{add}^{-1} \circ \text{add}^{-1} \circ \text{root})(2^x) = (\text{add}^{-1} \circ \text{add}^{-1} \circ \text{root} \circ \exp)(x).$

3. Induction and recursion (5+10 points)

Consider the sequence $(t_n)_{n \in \mathbb{N}}$ of numbers defined recursively by

$$t_0 := 1, \quad t_{n+1} := t_n + 2^n - 1.$$

We claim that the following statement is true for all natural numbers n :

$$t_n = 2^n - n.$$

- a) Verify by explicit computation that the statement is true for $n = 0$, $n = 1$, $n = 2$, and $n = 3$.

Solution: $t_0 = 1 = 2^0 - 0$; $t_1 = 1 + 2^0 - 1 = 1 = 2^1 - 1$; $t_2 = 1 + 2^1 - 1 = 2 = 2^2 - 2$;
 $t_3 = 2 + 2^2 - 1 = 5 = 2^3 - 3$.

- b) Prove by mathematical induction that the statement holds true for all natural numbers n .

Solution:

– *Base case:* $t_0 = 1 = 2^0 - 0$.

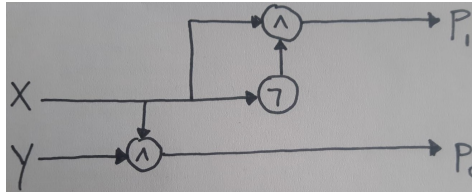
– *Induction step:* Let $m \geq 0$. Assume $t_m = 2^m - m$ (IH).

Then $t_{m+1} = t_m + 2^m - 1 \stackrel{\text{(IH)}}{=} 2^m - m + 2^m - 1 = 2 * 2^m - m - 1$ and
 $2^{m+1} - (m + 1) = 2 * 2^m - m - 1$.

Hence (IH) implies $t_{m+1} = 2^{m+1} - (m + 1)$.

4. Logic circuits (9 points)

Explain why the logic circuit below expresses the result $p_1 p_0$ of multiplying two bits x and y in binary arithmetic.



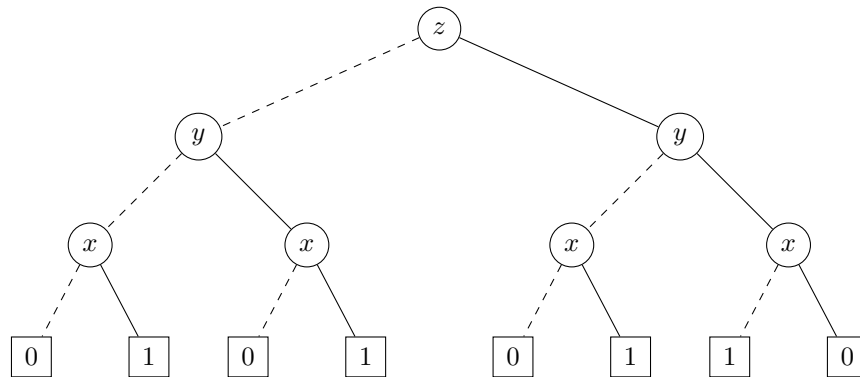
Solution: The product of x and y is 00 if x or y is 0, and is 01 if x and y are both 1. So p_1 is always 0; and p_0 is 0 if x or y is 0, and 1 if x and y are both 1.

In the logic circuit, p_1 is expressed as $x \wedge \neg x$, which is a contradiction so always 0. And p_0 is expressed as $x \wedge y$, so is 1 only if x and y are both 1.

5. OBDDs (4+10 points)

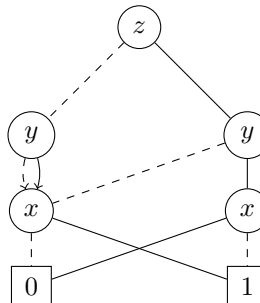
- (a) Represent $x \oplus (y \wedge z)$ by means of a binary decision tree, with respect to the variable ordering z, y, x .

Solution:

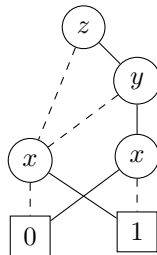


- (b) Reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

Solution: Apply C1 twice, to collapse leaves, yields



Apply C2 to remove the left-hand node y :



6. Syntax of predicate logic (8 points)

Use the predicates

$B(x)$: x is a book in the library

$M(x)$: x is a member of the library

$R(x, y)$: x read y

to translate the phrase “No member read every book in the library” into a predicate logic formula.

Solution: $\forall x (M(x) \rightarrow \exists y (B(y) \wedge \neg R(x, y)))$

7. Semantic of predicate logic (10+4 points)

Consider the formulas $\forall x (P(x) \rightarrow \exists x Q(x))$ and $\forall x (P(x) \rightarrow Q(x))$.

- a) Give a model to show that they are not semantically equivalent.

Solution: Consider a model $\mathcal{M}$ with $A = \{a, b\}$, and only $Q(a)$ and $P(b)$ hold. The left-hand side formula holds in $\mathcal{M}$, because Q holds for some element (namely a). But the right-hand side formula does not hold in $\mathcal{M}$, because $P(b)$ holds but $Q(b)$ does not hold in $\mathcal{M}$, and hence $P(b) \rightarrow Q(b)$ does not hold in $\mathcal{M}$.

- b) Explain which of these formulas semantically entails the other formula.

Solution: The right-hand side formula semantically entails the left-hand side formula. Consider any model for which the right-hand side formula holds. Then for all elements a in this model for which $P(a)$ holds, also $Q(a)$ holds. Hence, for all elements a for which $P(a)$ holds, there is an element b for which $Q(b)$ holds, namely $b = a$. This means that the left-hand side formula also holds in this model.